Bias Investigation Report

Organization: TalentMatch AI

System Under Review: HireScore — AI Applicant Ranking Tool

Investigator Role: QA Engineer

Date: 12 May 2026

1. Executive Summary

Overall Bias Severity: HIGH

HireScore exhibits systemic, multi-layered bias that significantly disadvantages female candidates, applicants from outside Greater Accra and Ashanti regions, and graduates of non-elite institutions. This bias is not a peripheral error but is structurally embedded throughout the machine learning pipeline—from the historical training data to the selection of proxy features and the resulting output scores.

Primary Affected Groups:

- **Female applicants:** Scored 1.4 points lower on average; representing only 11% of the top-100 ranked candidates.
- **Regional Applicants:** Candidates from Northern Ghana and other non-Accra/Ashanti regions represent only 9% of the top-100.
- **Institutional Diversity:** Graduates from institutions outside the University of Ghana, KNUST, and Ashesi represent only 6% of the top-100.

Top 3 Recommended Actions:

1. **Immediate Feature Removal:** Drop "University Name," "LinkedIn Connections," and "Previous Company Names" to eliminate proxy variables encoding socio-economic and regional privilege.
2. **Implementation of "Blind" Human Review:** Redact identifiers from the top 100 candidates to prevent human affinity bias from reinforcing algorithmic errors.
3. **Adoption of the Equal Opportunity Fairness Standard:** Shift the model's goal from "Demographic Parity" to "Equal Opportunity" to ensure qualified candidates have an equal chance of high scores regardless of gender or origin.

2. Detailed Findings

2.1 Bias Type Analysis

- **Historical Bias:** The training data relies on 10 years of past hiring decisions where 72% of successful hires were male. The model has "learned" that being male is a predictor of success, rather than a reflection of past societal barriers.
- **Representation Bias:** 85% of the training data comes from just two regions (Greater Accra and Ashanti). The model lacks the "experience" to accurately score qualified candidates from other regions, leading to their systematic exclusion.
- **Measurement Bias:** Features like "LinkedIn Connections Count" are used as proxies for "Professionalism." In reality, these measure social capital and age, disadvantaging younger candidates or those from less-networked regional backgrounds.

2.2 Pipeline Mapping

- **Data Collection:** Bias is introduced here via the "Ground Truth" problem—historical hires are treated as "ideal," ignoring the human biases that led to those hires.
- **Feature Engineering:** This is the most critical point of failure. Features such as "University Name" act as **Proxy Variables** for institutional prestige rather than actual skill.
- **Model Evaluation:** The current evaluation focuses on "Accuracy" (matching past hires) rather than "Fairness" (equitable distribution of scores).

3. Mitigation Plan

Action Item	Owner	Success Metric	Timeline
Immediate: Feature Redaction	Data Science	Correlation between University/Location and Score drops to <0.05.	Week 1

Short-term: Blind Human Review	Product/UX	>90% agreement between AI and Blind Human Reviewers.	Month 1-3
Short-term: Data Re-weighting	Engineering	HireScore achieves an Impact Ratio >0.8 (4/5ths Rule) for gender.	Month 3
Long-term: Equal Opportunity Framework	QA / Legal	Equal True Positive Rates across all protected groups.	Month 6-12

4. Ethical & Strategic Conclusion

HireScore is currently a high-severity case of compounded algorithmic bias. These flaws are not "bugs"—they are reflections of a design process that treated historical data as neutral truth.

The proposed shift toward **Equal Opportunity** is a business imperative. A tool that systematically screens out half of Ghana's qualified talent pool is not a competitive tool; it is a liability. By moving to a "Blind Review" model and removing proxy features, TalentMatch AI can pivot from encoding past injustice to supporting fair, skill-based hiring.

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Classification: Internal – Sensitive